

Week 9 Report: Results Analysis and Dissertation Writing (start)

Project: Determining the Role of Big Data Applications in Fraud Detection and Financial Security in Financial Institutions of Ireland

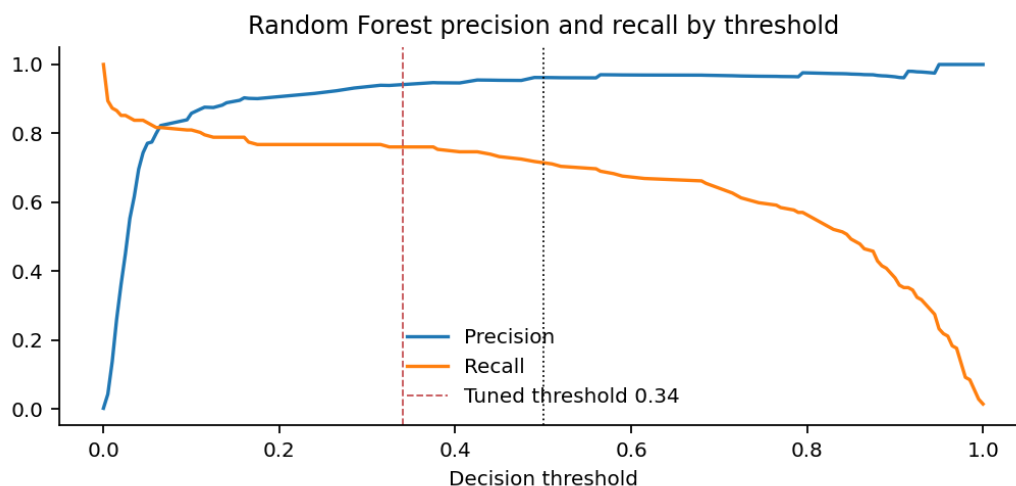
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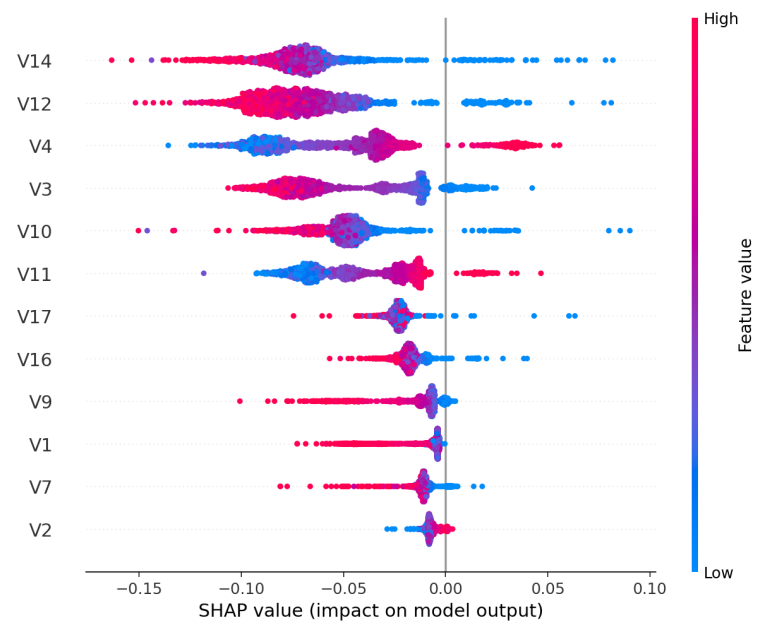
What I did this week

This week I analysed the results further. Choosing the threshold on validation data instead of using 0.5 raised Logistic Regression from F1 0.100 to 0.785, Decision Tree from 0.570 to 0.741, and Random Forest from 0.818 to 0.844.

When I trained on the first 70% of the time period and tested on the rest, Random Forest kept an F1 of 0.791, while Decision Tree fell to 0.414 and Logistic Regression to 0.081. The hour feature made no difference to Random Forest (F1 0.818 with and without it). Training on the full training set took 16.4 seconds. The most important features by permutation were V4, V9, V12, V3, V14. I also started the dissertation outline.



Random Forest precision and recall at different thresholds.



SHAP summary for Random Forest.

Next week

Write the introduction, literature review and methodology chapters.